

TP3 – Exercise 4: UML

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October 9, 2026

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1 Exercise Statement

A start-up plans to develop a hotel management application that can be used both on the web and on mobile devices. The goal of this system is to assist hotel staff and customers in managing reservations, stays, billing, and room maintenance efficiently.

A hotel client can make a reservation either by phone, through the web interface, or via the mobile app. When the estimated amount of the reservation exceeds 5000 DH, the client is required to make an online payment of at least 10% of the reservation amount using a CIB bank card. This payment is verified and validated by the client's bank server, with all communications secured through SSL.

Upon the client's arrival, the receptionist uses the application to search for available rooms that meet the client's preferences. The receptionist can also verify an existing reservation. Once a suitable room is identified, the receptionist performs the check-in operation: the room status changes from "available" to "occupied." At the same time, a stay record is created for the client.

The receptionist generates an invoice for the duration of the stay. During the stay, the client may request additional services such as breakfast, lunch, or dinner. The cashiers record these consumptions in the system, and the billing module automatically adds the corresponding charges to the client's invoice.

At the end of the stay, the receptionist performs the check-out procedure using the application. The receptionist closes the client's stay and completes the invoice after receiving payment. Once the client leaves, the room status changes to "dirty free."

Next, the housekeeping staff clean all rooms marked as "dirty free." After cleaning, they update the room's status to "clean awaiting." The floor manager then inspects the rooms to ensure they are properly cleaned. If everything is satisfactory, the manager changes the room's status to "available." However, if a room has any problems (such as a broken air conditioner or water leak), the floor manager marks the room as "unavailable." The room remains unavailable until maintenance staff carry out the necessary repairs, after which it becomes available again.

2 Questions

2.1 Question 1: Identify the Actors

Based on the text, identify the actors of the system.

The actors are:

1. Client
2. Client's bank server
3. Receptionist
4. Cashier
5. Housekeeping staff
6. Floor manager
7. Maintenance staff

2.2 Question 2: Actions Outside the System

Identify the actions that occur outside the system:

1. The client makes a reservation by phone.
2. The client's bank server verifies and validates the payment.
3. The client requests additional services.
4. The receptionist receives the client's payment and completes the invoice.
5. The housekeeping staff clean rooms marked as "dirty free."
6. The floor manager inspects the cleaned rooms.
7. The maintenance staff carry out the necessary repairs.

2.3 Question 3: System Boundaries

If we wanted to decompose the system into boundaries, what would these boundaries be?

The boundaries are:

1. Reservation Management
2. Payment Management
3. Stay Management
4. Billing Management
5. Housekeeping Management
6. Maintenance Management

2.4 Question 4: Use Case Diagram

Draw the use case diagram of the system with functional boundaries (modules).

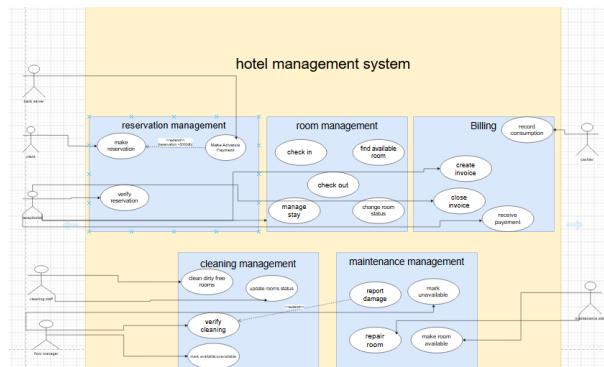


Figure 1: My diagram